



Multiplying and Dividing Complex Numbers in Trigonometric Form

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Learning Objectives

- Multiply complex numbers expressed in trigonometric (polar) form
- Divide complex numbers expressed in trigonometric (polar) form
- Simplify trigonometric form results and convert to standard $a+bi$ form when required

Perform the indicated operation on the complex numbers given in trigonometric form and simplify your answer.

1. Simplify the complex number into standard form $a+bi$.

$$3(\cos 45^\circ + i\sin 45^\circ)$$

Answer: _____

2. Simplify the complex number into standard form $a+bi$.

$$\sqrt{3}(\cos 390^\circ + i\sin 390^\circ)$$

Answer: _____

3. Let $z_1 = 4(\cos 60^\circ + i\sin 60^\circ)$ and $z_2 = 2(\cos 30^\circ + i\sin 30^\circ)$. Find $z_1 \cdot z_2$ in trigonometric form.

$$z_1 \cdot z_2 = 4(\cos 60^\circ + i\sin 60^\circ) \cdot 2(\cos 30^\circ + i\sin 30^\circ)$$

Answer: _____

4. Let $z_1 = 6(\cos 120^\circ + i\sin 120^\circ)$ and $z_2 = 3(\cos 45^\circ + i\sin 45^\circ)$. Find z_1/z_2 in trigonometric form.

$$\frac{z_1}{z_2} = \frac{6(\cos 120^\circ + i\sin 120^\circ)}{3(\cos 45^\circ + i\sin 45^\circ)}$$

Answer: _____

5. Multiply the complex numbers and express the result in trigonometric form.

$$5(\cos 80^\circ + i\sin 80^\circ) \cdot 2(\cos 100^\circ + i\sin 100^\circ)$$

Answer: _____

6. Divide the complex numbers and express the result in trigonometric form.

$$\frac{12(\cos 200^\circ + i\sin 200^\circ)}{4(\cos 110^\circ + i\sin 110^\circ)}$$

Answer: _____



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7. Multiply the complex numbers and simplify to standard form $a+bi$.

$$\sqrt{2}(\cos 135^\circ + i\sin 135^\circ) \cdot 3(\cos 45^\circ + i\sin 45^\circ)$$

Answer: _____

8. Divide the complex numbers and simplify to standard form $a+bi$.

$$\frac{8(\cos 210^\circ + i\sin 210^\circ)}{2(\cos 60^\circ + i\sin 60^\circ)}$$

Answer: _____

9. Find the product $(z_1)(z_2)$ in trigonometric form, where $z_1 = 7(\cos 25^\circ + i \sin 25^\circ)$ and $z_2 = 4(\cos 50^\circ + i \sin 50^\circ)$.

$$z_1 \cdot z_2$$

Answer: _____

10. Find z_1/z_2 in trigonometric form, where $z_1 = 10(\cos 300^\circ + i \sin 300^\circ)$ and $z_2 = 5(\cos 150^\circ + i \sin 150^\circ)$.

$$\frac{z_1}{z_2}$$

Answer: _____





Multiplying and Dividing Complex Numbers in Trigonometric Form

ANSWER KEY & SOLUTIONS

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Remind students that when multiplying, moduli multiply and arguments add; when dividing, moduli divide and arguments subtract.

Solutions

1. Simplify the complex number into standard form $a+bi$.

$$3(\cos 45^\circ + i \sin 45^\circ)$$

→ Substitute $\cos 45^\circ = \sqrt{2}/2$ and $\sin 45^\circ = \sqrt{2}/2$.

→ Distribute the 3 to each term to get $3(\sqrt{2}/2) + 3(\sqrt{2}/2)i$.

→ Write the result as $(3\sqrt{2})/2 + (3\sqrt{2})/2 i$.

Answer: $\frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i$

2. Simplify the complex number into standard form $a+bi$.

$$\sqrt{3}(\cos 390^\circ + i \sin 390^\circ)$$

→ Reduce 390° by subtracting 360° to get a coterminal angle of 30° .

→ Substitute $\cos 30^\circ = \sqrt{3}/2$ and $\sin 30^\circ = 1/2$.

→ Multiply by $\sqrt{3}$ to get $(\sqrt{3})(\sqrt{3}/2) + (\sqrt{3})(1/2)i = 3/2 + (\sqrt{3}/2)i$.

Answer: $\frac{3}{2} + \frac{\sqrt{3}}{2}i$

3. Let $z_1 = 4(\cos 60^\circ + i \sin 60^\circ)$ and $z_2 = 2(\cos 30^\circ + i \sin 30^\circ)$. Find $z_1 \cdot z_2$ in trigonometric form.

$$z_1 \cdot z_2 = 4(\cos 60^\circ + i \sin 60^\circ) \cdot 2(\cos 30^\circ + i \sin 30^\circ)$$

→ Multiply the moduli: $4 \times 2 = 8$.

→ Add the arguments: $60^\circ + 30^\circ = 90^\circ$.

→ Write the product as $8(\cos 90^\circ + i \sin 90^\circ)$.

Answer: $8(\cos 90^\circ + i \sin 90^\circ)$

4. Let $z_1 = 6(\cos 120^\circ + i \sin 120^\circ)$ and $z_2 = 3(\cos 45^\circ + i \sin 45^\circ)$. Find z_1/z_2 in trigonometric form.

$$\frac{z_1}{z_2} = \frac{6(\cos 120^\circ + i \sin 120^\circ)}{3(\cos 45^\circ + i \sin 45^\circ)}$$

→ Divide the moduli: $6 \div 3 = 2$.

→ Subtract the arguments: $120^\circ - 45^\circ = 75^\circ$.

→ Write the quotient as $2(\cos 75^\circ + i \sin 75^\circ)$.

Answer: $2(\cos 75^\circ + i \sin 75^\circ)$



5. Multiply the complex numbers and express the result in trigonometric form.

$$5(\cos 80^\circ + i \sin 80^\circ) \cdot 2(\cos 100^\circ + i \sin 100^\circ)$$

→ Multiply the moduli: $5 \times 2 = 10$.

→ Add the arguments: $80^\circ + 100^\circ = 180^\circ$.

→ Write the product as $10(\cos 180^\circ + i \sin 180^\circ)$, which simplifies to -10 .

Answer: $10(\cos 180^\circ + i \sin 180^\circ)$

6. Divide the complex numbers and express the result in trigonometric form.

$$\frac{12(\cos 200^\circ + i \sin 200^\circ)}{4(\cos 110^\circ + i \sin 110^\circ)}$$

→ Divide the moduli: $12 \div 4 = 3$.

→ Subtract the arguments: $200^\circ - 110^\circ = 90^\circ$.

→ Write the quotient as $3(\cos 90^\circ + i \sin 90^\circ)$, which simplifies to $3i$.

Answer: $3(\cos 90^\circ + i \sin 90^\circ)$

7. Multiply the complex numbers and simplify to standard form $a+bi$.

$$\sqrt{2}(\cos 135^\circ + i \sin 135^\circ) \cdot 3(\cos 45^\circ + i \sin 45^\circ)$$

→ Multiply the moduli: $\sqrt{2} \times 3 = 3\sqrt{2}$.

→ Add the arguments: $135^\circ + 45^\circ = 180^\circ$.

→ Evaluate: $3\sqrt{2}(\cos 180^\circ + i \sin 180^\circ) = 3\sqrt{2}(-1 + 0i) = -3\sqrt{2}$.

Answer: $-3\sqrt{2}$

8. Divide the complex numbers and simplify to standard form $a+bi$.

$$\frac{8(\cos 210^\circ + i \sin 210^\circ)}{2(\cos 60^\circ + i \sin 60^\circ)}$$

→ Divide the moduli: $8 \div 2 = 4$.

→ Subtract the arguments: $210^\circ - 60^\circ = 150^\circ$.

→ Substitute $\cos 150^\circ = -\sqrt{3}/2$ and $\sin 150^\circ = 1/2$; multiply by 4 to get $-2\sqrt{3} + 2i$... correct: $4(-\sqrt{3}/2) + 4(1/2)i = -2\sqrt{3} + 2i$.

Answer: $-2\sqrt{3} - 2i$

9. Find the product $(z_1)(z_2)$ in trigonometric form, where $z_1 = 7(\cos 25^\circ + i \sin 25^\circ)$ and $z_2 = 4(\cos 50^\circ + i \sin 50^\circ)$.

$$z_1 \cdot z_2$$

→ Multiply the moduli: $7 \times 4 = 28$.

→ Add the arguments: $25^\circ + 50^\circ = 75^\circ$.

→ Write the product as $28(\cos 75^\circ + i \sin 75^\circ)$.

Answer: $28(\cos 75^\circ + i \sin 75^\circ)$

10. Find z_1/z_2 in trigonometric form, where $z_1 = 10(\cos 300^\circ + i \sin 300^\circ)$ and $z_2 = 5(\cos 150^\circ + i \sin 150^\circ)$.

$$\frac{z_1}{z_2}$$

→ Divide the moduli: $10 \div 5 = 2$.

→ Subtract the arguments: $300^\circ - 150^\circ = 150^\circ$.

→ Write the quotient as $2(\cos 150^\circ + i \sin 150^\circ)$.

Answer: $2(\cos 150^\circ + i \sin 150^\circ)$

